

Consultant's comments reply:

1. Provide point by point compliance by stating "*Comply*" or "*Not comply*" with technical reason to specification # 262419 such as, clauses # 2.1.I.1 to 2.1.I.12, 2.1.C, 2.1.H.

Reply: Detailed compliance sheet with each point (Comply or Not Comply available) with detailed technical reason

2. Provide missing AC line reactors and DC buffers for each offered VFDs as required in clauses # 2.1.D & 2.1.E.

Reply: For this point we advised that ATV600 Embedded low harmonic DC choke technology complying with standard IEC 61000-3-12 for the 380...480 V offer to reach THDi less than 48% for 80 to 100% load, which is used to maintain an optimum power factor on the most common operating range

3. Manufacturer shall fully comply with specification # 262419 clause # 2.1.J.9 (Not option).

Reply: Our ATV650 product has a built in 3 relay outputs 1 with NO/NC contacts and 2 with NO contacts, For the point 2.1.J.9 VFD to have terminal strip and N.C. safety contacts such as freezstats, smoke alarms it will be programming simply by one of those 3 relay outputs without any problem .

4. Manufacturer shall comply to all general & VFDs related clauses in specification # 262419 such as clauses # 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.10. and any clause in the specification that is contractor scope shall be complied by contractor. In addition, provide missing compliance to some clauses of specification # 262419 such as clause # 2.1.G.

Reply: For Clause #2.1.G we acknowledge that we have a built in function to reduce the motor noise level as per the programming manual limited to the required number by specs .

For Clause 1.1 VSD drawings are available as per the attached document

For Clause 1.2 Its related to the MCCs & BMS panels and not replated to VSD as it's not located in side the panel and selected as a wall mounted with IP55

For Clause 1.3 Seismic Performance is not available with the VSD

For Clause 1.4 All the technical details about any drive are available on the data sheets (attached per each reference) also the dimensional drawings . Full compliance sheet is attached

For Clause 1.5 All VSD are made by Schneider Electric Factory and all the required IEC compliance are available as follow:

Standards EN/IEC 61800-3 EN/IEC 61800-3 environment 1 category C2 EN/IEC 61800-3 environment 2 category C3 EN/IEC 61800-5-1 IEC 61000-3-12 IEC 60721-3 IEC 61508 IEC 13849-1 UL 508C

For Clause 1.6 Standard Warranty period is 12 months from commissioning or 18 Month from delivery and to reach 2 years from commissioning it will be with extra cost (optional)

For Clause 1.7 For the VSD It will be the contractor scope and it will be as per recommended on the data sheet per each drive Ambient air temperature for storage -40...70 °C

For Clause 1.8 All the required coordination from our side with the contractor will be available during the order stage

For Clause 1.9 For all the VSDs recommended spare part is available but it will be a complete drive unit to cover most of the ranges

For Clause 1.1 FAT, The available test is routine test as per the attached one

5. The submitted list of material from page 307 to 316 shall include type and part no. of each offered item and is incomplete as it is missing main items such as:

- VFDs types & ratings,
- Protective devices of VFDs (Fuses, CBs, etc.),
- All accessories (Such as, indication lamps & selectors).
- Busbars data such as, material / ratings / short circuit / insulation / plating / etc.
- All other materials that will be installed in enclosure.

All to be supported by highlighted data sheet noting that VFDs data sheets shall be highlighted on the selected models.

Reply: As per the ATV Catalogue the selection of the Contactors, CBs , Fuses are available

As attached

6. Provide / coordinate with BMS and any other interface required in specification and design drawings.

Reply: Our ATV600 drive has the below built in I/O's & Communication modules

Analog inputs 3: Configurable as voltage (0...10 V) or current (0-20 mA/4-20 mA), 2 of them including probes

(PTC, PT100, PT1000, or KTY84)

Digital inputs 6: Voltage 24 V c (positive or negative logic)

Analog outputs 2: Configurable as voltage (0...10 V) or current (0-20 mA)

Relay outputs 3: 1 with NO/NC contacts and 2 with NO contacts

Safety function inputs 2: For safety function STO

Modbus serial, Modbus TCP

drawings.

7. Contractor / manufacturer to confirm / ensure that all indicated motors ratings / means of starting / selection of all VFDs ratings / etc. in the subject submittal are as per approved mechanical submittals and shall be coordinated with the mechanical equipment suppliers. In addition, indicate all selected VFDs are normal or heavy duties in coordination with mechanical submittals.

Reply: Attached the Motor power, Selected VSD at 50 degree C as per specs, data sheet for each VSD that indicate the nominal power on Normal & Heavy-duty conditions

8. All VFDs circuit components (VFDs, fuses, CBs, contactors, etc.) shall be based on Schneider type 2 coordination tables at derated ambient of 50°C, manufacturer to submit these Schneider tables for our cross check. In addition, revise your selection for contactors ratings (For example only, 25A contactors selected for 22KW/40A VFDs is downsized and shall be upgraded to suit VFDs ratings, etc.).

Reply: As per the attached Motor power, Selected VSD at 50 degree C as per specs, data sheet for each VSD that indicate the nominal power on Normal & Heavy-duty conditions, Also as per the attached document from the ATV600 catalogue about the selection of the motor starts like contactor

22 KW drive ATV650D22N4 the selected contactor will be LC1D50A

18.5	25	ATV650D18N4/N4E	GV3L40	40	550	LC1D40A●●
22	30	ATV650D22N4/N4E	GV3L50	50	700	LC1D50A●●
30	40	ATV650D30N4/N4E	GV3L65	65	810	

9. The submitted list of material from page 307 to 316 missing to include complete enclosure type and dimensions and missing enclosure highlighted technical data sheet and type test certificate, manufacturer to submit.

Reply: For VSD the available are the routine test done on our factory (attached)

10. Submit reference approved design & shop drawings (Single line diagrams, etc.) of the offered panels & their upstream feeding panels for our cross reference.

Reply: Not related to VSD scope

11. Indicate by supporting document the location of installation of these panels (Indoor electrical substation or mechanical rooms or wet areas or what ?, outdoors).

Reply: for the selected VSD they are wall mounted with IP55 , So they are ready to be located outdoor

**12. Submit complete manufacturer factory panels shop drawings including but not limited to:
with/without door dimensional plans & front and side views, general arrangement, wiring**

Reply: Not related to VSD scope